Helix Forge Research Method Validation Methods Study

HELIX FORGE LAB / p04-full-006599 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded method validation evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000055 (research/papers/2026-07-13/helix-forge-research/failure-analysis/tables/helix-forge-research-failure-analysis-thresholds-v01.csv).

Slice	Records	Precision	Recall	F1
slice-01	225	0.872	0.865	0.868
slice-02	182	0.889	0.878	0.883
slice-03	179	0.871	0.854	0.862
slice-04	243	0.888	0.867	0.877
slice-05	199	0.870	0.880	0.875